

SURFACE-WATER-QUALITY DYNAMICS IN THE MEKONG DELTA UNDER POLLUTION PRESSURES AND ALTERED FLOW REGIMES

Vu Thi Hoai Thu^(1,2*)

⁽¹⁾University of Transport Ho Chi Minh City, Ho Chi Minh City, Viet Nam

⁽²⁾Institute of Water Resources and Environment Technology, Viet Nam

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Abstract: Surface-water quality in the Vietnamese Mekong Delta is deteriorating under combined pollution pressures and altered hydrological conditions. This study synthesizes 2016-2024 monitoring records, a 3,117-sample water-quality database, supplementary surveys conducted in 2022-2024, source-apportionment outputs, and hydrodynamic-salinity-morphological evidence to clarify how pollution loads and changes in assimilative capacity jointly shape water quality index (WQI) dynamics. WQI ranges from 2.376 to 85.314, with a regional mean of 47.486, indicating widespread fair-to-poor conditions in coastal, peri-urban, aquaculture-intensive, and poorly flushed canal areas. Principal component analysis (PCA), factor analysis (FA), and agglomerative hierarchical clustering (AHC) identify five pollution components: organic wastewater, agricultural runoff and pesticides, microbial contamination associated with Zn/Cu, salinity-related mobilization of As and Cr, and combined nutrient-microbial pollution. Source apportionment shows domestic/industrial dominance in urban and estuarine corridors, while agriculture and aquaculture are more influential in inner-delta and coastal settings. The main scientific contribution is an integrated mechanism showing that altered flow regimes do not generate pollutants directly but reduce dilution, increase residence time, deepen salinity intrusion, and lower assimilative capacity, thereby amplifying water-quality decline.

Keywords: Surface water quality, Mekong Delta, pollution sources, altered flow, salinity intrusion.

1. Introduction

The Mekong Delta is the terminal downstream reach of the Mekong system and, at the same time, a major space for agriculture, aquaculture, and settlement whose functioning depends heavily on surface water. Over roughly the last decade, surface water quality in the delta has shown a deteriorating tendency in both space and time, as reflected by a regional mean WQI in the fair-to-poor range; many peri-urban, coastal, and inner-canal locations frequently exhibit elevated BOD₅, COD, NH₄⁺, and Coliform concentrations [1], [2]. This degradation trend increases the risk that water-quality decline will become a chronic constraint on freshwater supply, aquaculture, irrigation, and ecosystem health rather than an isolated seasonal problem.

Notably contemporary water-quality variability in the Mekong Delta cannot be explained solely by the growth of pollution loads. Upstream flow alteration, declining sediment supply, channel-bed lowering, deeper salinity intrusion, and the operation of sluices, dikes, and inner-delta canals have changed dilution conditions, water residence time, and material transport-deposition mechanisms across the system. In this paper, “altered flow regimes” is used consistently to refer to changes in discharge timing, water level, salinity/tidal propagation, residence time, and hydraulic-structure operation; “assimilative capacity” refers to the ability of the river-canal system to dilute, transform, retain, or export pollutant loads without unacceptable WQI deterioration. In other words, pollution sources are the direct drivers, whereas altered flow regimes amplify

Corresponding author: Vu Thi Hoai Thu
E-mail: hoaitu.vu@ut.edu.vn

water-quality risk by reducing assimilative capacity [3-6], [8], [9].

In many earlier studies, these two issue groups were often examined separately: either with a focus on pollution-status assessment, or with emphasis on salinity intrusion and hydrological variability. This paper adopts an integrated perspective, combining water-quality data, pollution-source apportionment, and evidence of flow alteration to clarify three questions: (i) The spatial and temporal dynamics of surface water quality; (ii) The impacts of major pollution-source groups; and (iii) The amplifying role of altered flows in the degradation of surface water quality across the Mekong Delta [1], [2], [7].

2. Study area and methods

2.1. Study area

The core analytical domain of this paper is the Vietnamese Mekong Delta (VMD), including the Tien and Hau river distributaries, inner canal networks, coastal estuaries, and representative

irrigation systems that control water exchange within the delta. Figure 1 deliberately retains the adjacent lower Dong Nai-Sai Gon/Dau Tieng-Tri An system and the Vung Tau-Kien Giang coastal receiving corridor because the project database and pollutant-transport context include source-receiving linkages between these connected Southern Vietnamese waters. This adjacent area is used only as hydrological context, monitoring/source-apportionment boundary, and coastal receiving-water linkage; it is not treated as part of the Mekong Delta in the VMD subregional WQI statistics. The VMD subregions interpreted in the paper include the Ca Mau Peninsula, the Long Xuyen Quadrangle, the interfluvial zone between the Tien and Hau rivers, the left bank of the Tien River, and the key irrigation systems Cai Lon-Cai Be, Quan Lo-Phung Hiep, and Tu Giac Long Xuyen [1], [2], [6], [7], [13]. Historical administrative names are retained where they are original labels of the monitoring datasets, while the scientific interpretation is organized by hydrological subregion and water-exchange system.

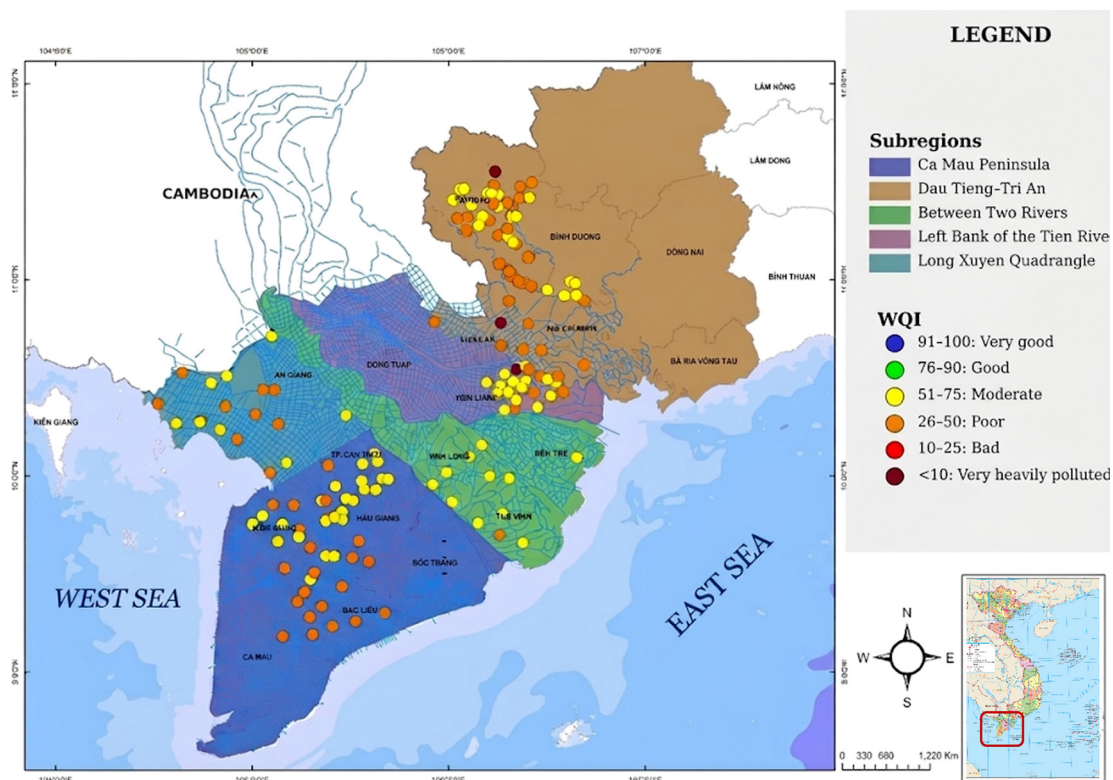


Figure 1. Study area, contextual hydrological linkage, spatial subdivision, and surface-water-quality monitoring network for the Vietnamese Mekong Delta and adjacent lower Dong Nai-Sai Gon/coastal receiving waters

The analytical database comprises: (i) Monitoring records for 2016-2024 including WQI, pH, DO, BOD₅, COD, NH₄⁺, NO₂⁻, NO₃⁻, PO₄³⁻, Coliform, and salinity; (ii) A dataset of 3,117 samples used for descriptive statistics, PCA, FA, and AHC; (iii) Supplementary surveys conducted during 2022-2024 for river, estuarine, nearshore, and adjacent receiving waters from Vung Tau to Kien Giang; (iv) Discharge-source inventory data and source-apportionment results from ESAT/MixSIAR; and (v) Hydrodynamic-salinity-morphological studies related to salinity intrusion, channel-bed incision, estuarine change, and the lower Dong Nai-Sai Gon coastal linkage [1], [3]-[13]. For clarity, VMD WQI statistics are calculated and interpreted for VMD stations and hydrological subregions; data from the lower Dong Nai-Sai Gon system are used only to support regional context and coastal source-receiving interpretation. No new concentration values were created in this revision; all numerical results were retained from the accepted monitoring, published, or project datasets.

2.2. Methods

The study applies a multi-source integrative approach. First, water-quality variables were summarized using ranges, mean values, and WQI-based use categories to identify high-risk areas and periods. Before PCA/FA/AHC synthesis, variables were standardized in the source study to make parameters with different units comparable; the five retained pollution components and spatial clusters are interpreted from the established 3,117-sample dataset rather than recalculated from new data in this revision [1]. Source-apportionment results from ESAT/MixSIAR were then used to explain the relative contributions of pollution-source groups by subregion and season. Finally, the water-quality results were interpreted together with changes in discharge, water level, salinity intrusion, channel morphology, and hydraulic-structure operation to elucidate the mechanisms that amplify risk [3-13].

This approach does not aim to build a new predictive model; rather, it links existing quantitative evidence into a unified interpretive

framework. The novelty of the paper lies in viewing surface-water-quality dynamics not merely as the result of in situ pollution loads, but as the combined outcome of pollution loads and changing assimilative capacity under altered flow regimes. This distinction is important because the same pollutant load may lead to different WQI outcomes depending on dilution, residence time, salinity, sediment remobilization, and hydraulic connectivity.

3. Results and discussion

3.1. Spatial and temporal dynamics of water quality

Synthesized results indicate that surface water quality in the Mekong Delta is at a concerning level when assessed across the region as a whole. WQI ranges from 2.376 to 85.314, with a mean value of 47.486; many locations are suitable only for inland navigation, or would require strict pollution control to meet higher use objectives [1]. The very wide ranges of BOD₅ and COD indicate the coexistence of reaches with some remaining self-purification capacity and hotspots receiving intense organic loads. Nutrient and microbial pollution are also widespread, especially in inner canals, peri-urban zones, and areas of intensive aquaculture.

The spatial differentiation among subregions is pronounced. The Ca Mau Peninsula generally exhibits low WQI because it is simultaneously affected by salinity pressure, intensive aquaculture, and limited water exchange; the Quan Lo-Phung Hiep system is particularly sensitive to organic and microbial pollution; whereas the Cai Lon-Cai Be system has shown improvement since salinity-control works became operational, although the recovery remains spatially uneven. The Long Xuyen Quadrangle is generally more stable, but the Western coastal belt still faces dry-season water-quality deterioration due to water stagnation and acid-sulfate/salinity effects [2], [7].

Temporally, the rainy and dry seasons produce two contrasting water-quality patterns. The rainy season increases the wash-off of nutrients, sediments, and pollutants from the land surface, causing strong event-scale variability in water

quality. In contrast, the dry season reduces dilution capacity, lengthens water residence time, raises the risk of local oxygen depletion, and increases BOD₅, COD, NH₄⁺, and Coliform in

poorly flushed canal reaches. This indicates that water-quality dynamics in the Mekong Delta are fundamentally seasonal, superimposed on a system already under high pollution pressure.

Table 1. Summary statistics of representative surface-water-quality variables in the Mekong Delta

Parameter	Range	Mean	Management implication
WQI	2.376 - 85.314	47.486	Many locations remain in the poor-to-fair range; pollution-source control and improved water exchange are needed.
DO (mg/L)	2.135 - 15.200	5.780	Some locations approach ecologically low thresholds; there is a risk of local oxygen depletion in the dry season.
BOD ₅ (mg/L)	3.500 - 172.870	10.627	Reflects high organic loads from domestic sources, industry, and intensive aquaculture.
COD (mg/L)	6.493 - 472.984	25.014	Indicates organic-pollution hotspots in inner-delta canals and peri-urban areas.
NH ₄ ⁺ (mg/L)	0.046 - 2.786	0.489	Sensitive to domestic wastewater, livestock waste, and weak hydrodynamic exchange.
PO ₄ ³⁻ (mg/L)	n.d. - 45.800	1.585	Risk of local eutrophication in canals receiving agricultural and aquaculture loads.
Coliform (MPN/100 mL)	n.d. - 45,100	4,408	Microbial contamination remains common across many subregions, especially densely populated areas.
As (mg/L)	0.006 - 0.046	-	May increase under saline and fluctuating redox conditions in estuarine zones.
Cr (mg/L)	n.d. - 1.960	-	Localized risk associated with industrial sources, inland navigation, and salinity-driven remobilization.

Note: n.d. = not detected

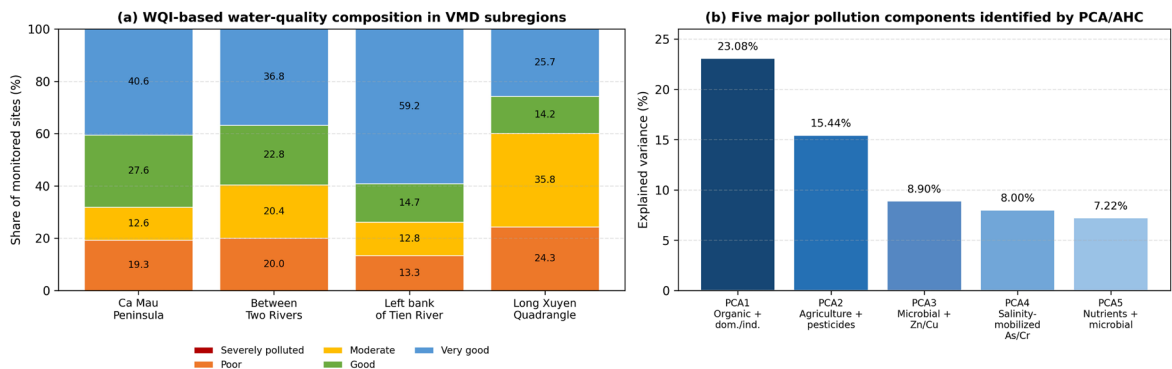


Figure 2. (a) WQI-based water-quality composition in representative VMD subregions; (b) five major pollution components identified by PCA/AHC

3.2. Effects of pollution sources on water quality

PCA/AHC analysis shows that surface-water pollution in the Mekong Delta is no longer a disconnected set of individual

indicators, but is structured into five relatively stable components [1]. The first component represents organic pollution associated with domestic and industrial wastewater; the second

reflects agricultural inputs and pesticide use; the third is related to microbial contamination associated with Zn/Cu; the fourth comprises salinity-driven mobilization of As and Cr; and the

fifth represents combined nutrient-microbial pollution. This structure makes it possible to identify pollution origins by mechanism rather than by isolated parameters alone.

Table 2. Dominant pollution-source contributions by subregion and season

Subregion/site	Wet season	Dry season	Control implication
Tan Chau - Chau Doc	Domestic 70-73%; agriculture 25-29%; industry 1-2%	Agriculture declines to 12-29%; industry increases slightly; domestic wastewater remains dominant	Prioritize domestic wastewater collection and treatment and reduce early-season wash-off.
Can Tho - Soc Trang (Hau River)	Industry 37-64%; domestic 24-32%; agriculture 12-31%	Can Tho: agriculture rises to 61%; Soc Trang: industrial share increases, domestic around 32%	Simultaneously control urban-industrial sources and aquaculture-pond effluents along the river.
Ben Tre - Tra Vinh (Tien River/estuary)	Industry 17-39%; agriculture 21-22%; domestic 40-99%	Industry increases by 11-22%; agriculture by 3-15%; domestic share varies strongly	Protect water-intake corridors, intensify dry-season monitoring, and control estuarine discharge points.
Co Chien estuary-Ca Mau Cape coastal sector	Agriculture 26-42%; atmospheric deposition 43-97%; other sources <20%	Some estuaries show sharp increases in industrial contribution; Dinh An - Ganh Hao are affected by both atmospheric deposition and agriculture	Monitor atmospheric deposition, tides, and estuarine mixing/ disturbance.
Ong Doc - Rach Gia	Ong Doc: domestic 52%, atmospheric deposition 48%; Rach Gia: domestic 55%, agriculture 36%	Ong Doc: atmospheric deposition 48%, domestic 27%, agriculture 25%; Rach Gia: atmospheric deposition 69%, domestic 31%	Strictly manage the river-sea interface and shrimp-farming zones near estuary mouths.

Note: Ca Mau Cape denotes the coastal endpoint of the Co Chien-Ca Mau sector in this table; Ca Mau Peninsula denotes the wider hydrological subregion discussed in the text.

Source-apportionment data show that the influence of pollution sources varies strongly by subregion and season. In the upstream Tan Chau-Chau Doc sector, domestic wastewater dominates during the rainy season (70-73%), while agriculture contributes 25-29%. Along the Can Tho-Soc Trang reach of the Hau River, industrial and urban sources dominate in the rainy season (37-64%), but the role of agriculture and aquaculture ponds increases markedly in the dry season. The Ben Tre-Tra Vinh corridor and coastal estuaries are simultaneously affected by domestic, industrial, agricultural, and atmospheric-deposition sources, whereas the Ong Doc-Rach Gia sector clearly reflects the

river-sea interface, with domestic, atmospheric, and aquaculture sources all contributing [7]. Fundamentally, pollution sources directly determine the initial load entering the surface-water system. Domestic and urban wastewater increase BOD₅, COD, NH₄⁺, and Coliform; agriculture and aquaculture increase nutrient, organic, and chemical-residue loads; industry adds refractory organic compounds and metals; and atmospheric deposition together with the river-sea interface give some coastal estuaries pollution characteristics that differ from those of inner-delta areas. Consequently, even under the same flow condition, different source structures produce very different water-quality outcomes.

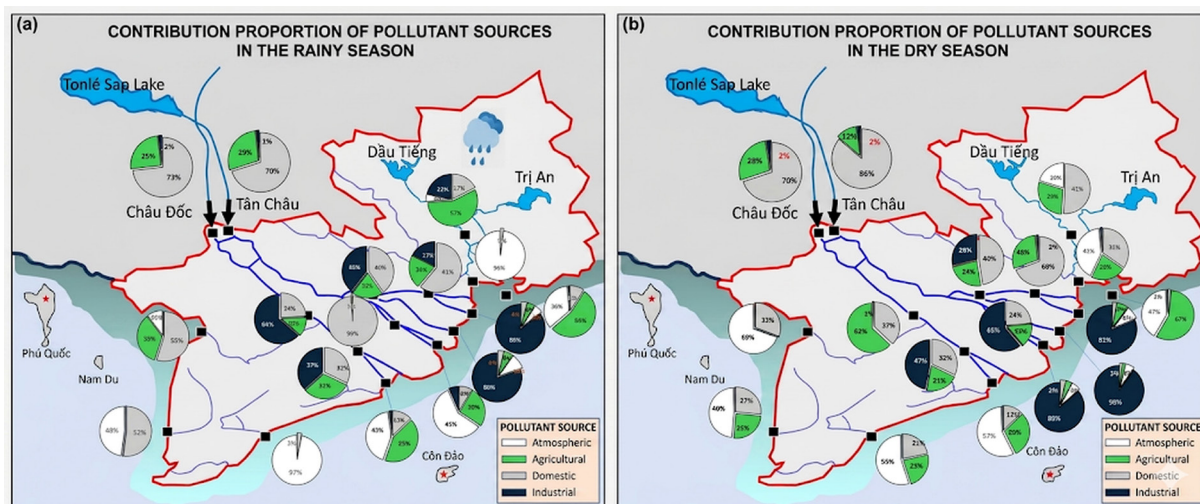


Figure 3. Seasonal contributions of major pollution sources across the interregional monitoring and source-apportionment space from the lower Dong Nai-Sai Gon/coastal receiving corridor to the Mekong Delta and Kien Giang

3.3. Amplifying role of altered flow regimes

If pollution sources are the direct cause, then altered flow regimes are the mechanism that regulates self-purification capacity and amplifies the impact of those sources. During the dry season, reduced low flows diminish dilution, increase water residence time in rivers and canals, and create conditions favorable for the accumulation of organic matter and nutrients. At the same time, deeper salinity intrusion changes the geochemical balance of the system, potentially mobilizing additional metals from sediments and increasing the risk of secondary pollution [4], [7].

A clear indicator of this mechanism is observed in the Soc Trang area, where dry-season surface water may contain up to about 29.5% seawater, whereas the rainy-season proportion is only about 4.12% [4]. This mixing not only increases salinity, but also alters the dissolution-adsorption equilibrium of substances in water and sediments. As salinity rises, water-supply and production-related quality indicators become more sensitive, especially at water-intake points, rice-shrimp farming zones, and freshwater conveyance canals into the inner delta.

Flow alteration in the Mekong Delta is also

linked to channel-bed lowering. In the Tien River, post-dam dry-season discharge may increase by about 23%, yet the low-flow water level at My Thuan has dropped substantially; this shows that increased discharge does not necessarily translate into the expected improvement in dilution when channel morphology has already changed [3]. The acceleration of bed incision, together with sediment depletion and intensified sand mining, has deepened the salinity boundary, prolonged the residence time of brackish and saline water, and made downstream areas more sensitive to a given pollution load [3], [8], [9], [11].

In addition, the operation of salinity-control structures has two-sided effects on water quality. Systems such as Cai Lon-Cai Be can control salinity over a large area and help create freshwater supplies for production, but if operated without explicitly integrating water-quality objectives, they may also alter residence time, flow distribution, and pollution accumulation zones within the inner delta [5]. Therefore, hydraulic operation should be evaluated not only by salinity targets but also by DO, NH_4^+ , Coliform, BOD_5/COD , and WQI responses at intake and drainage corridors.

Table 3. Hydrological, salinity, and morphological indicators directly related to water-quality dynamics

Indicator	Key value	Link to water quality	Practical implication
Seawater mixing in surface water	29.5% in the dry season; 4.12% in the wet season (Soc Trang area)	Increases salinity, shifts dissolution-adsorption equilibria, and promotes the mobilization of As and Cr	Protect water-intake points and operate sluices in real time during the dry season.
Low-flow discharge of the Tien River	Increases by about 23% after dam construction	Higher discharge does not necessarily improve water quality when the base water level continues to fall	Evaluate discharge, water level, salinity, and pollution sources jointly.
Low-flow water level at My Thuan	Decreases by about 46%	Reduces the water-surface gradient, drives the salinity boundary farther inland, and weakens dilution capacity	Strengthen inter-reservoir and interregional operation with explicit water-quality objectives.
Channel-bed lowering	From about -0.16 m/year to nearly -0.50 m/year	Intensifies salinity intrusion, prolongs brackish-saline residence time, and increases the risk of DO decline	Reduce sand mining and monitor channel morphology regularly.
Sediment load / sand mining	166.7 → 57.6 Mt/year; 3.9 → 13.43 Mm ³	Sediment depletion weakens morphological recovery capacity and increases erosion and pollutant remobilization	Strengthen basin-scale management and tighten sand-mining control in main channels.
Operation of salinity-control sluices	Controls about 400,000 ha and creates additional water sources for about 500,000 ha	Can improve water availability but may also alter residence time and pollution distribution if operated asynchronously	Integrate fresh-saline objectives with water-quality and ecological objectives.

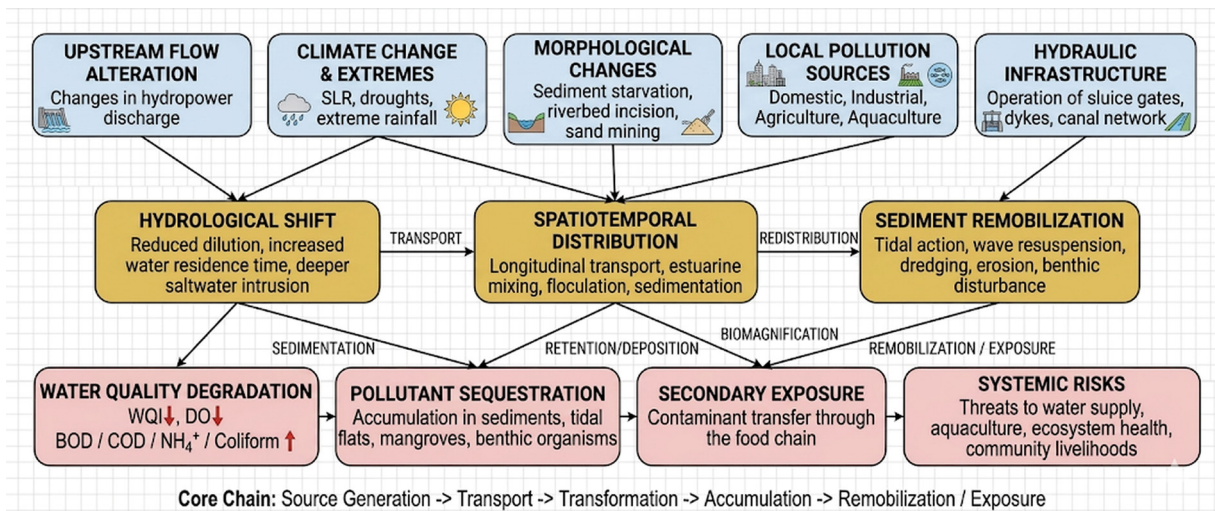


Figure 4. Integrated conceptual mechanism linking pollution loads, altered flow regimes, reduced assimilative capacity, pollutant remobilization, and WQI decline. The mechanism summarizes evidence from VMD water-quality monitoring, salinity intrusion, channel-bed incision, hydraulic-structure operation, and adjacent coastal source-receiving linkages [1]–[13].

3.4. Management implications

The results indicate that surface-water-quality management in the Mekong Delta must move from a single-variable mindset toward integrated management of pollution sources, flows, salinity, and hydraulic infrastructure. Along urban corridors and in estuarine zones, the highest priority is the control of domestic wastewater, industrial effluents, and aquaculture-pond discharge before they enter rivers. In inner-delta areas, agricultural wash-off should be reduced through improved fertilizer and pesticide management, ecological buffer zones, and settling ponds. By subregion, priority should differ: the Ca Mau Peninsula and Quan Lo-Phung Hiep require stronger aquaculture/domestic-load control and controlled flushing; Cai Lon-Cai Be requires operation rules that combine salinity control with WQI, DO, NH_4^+ , and Coliform thresholds; Tu Giac Long Xuyen and the Western coastal belt require measures to reduce stagnation, acid-sulfate/salinity effects, and dry-season oxygen stress; and the Tien-Hau urban-estuarine corridors require strict protection of intake zones and wastewater-treatment performance.

In operational terms, WQI should be interpreted together with discharge, water level, salinity, DO, NH_4^+ , and Coliform. During the dry season, the operation of intake sluices, freshwater retention, or water-exchange releases must be linked to real-time water-quality information; if management relies only on fresh-saline objectives, the risk of importing poor-quality water into the inner delta remains high. Therefore, high-frequency monitoring networks should be strengthened and linked with WebGIS, data assimilation, and early-warning systems to support decision-making [2], [7].

In the longer term, sediment depletion, channel-bed lowering, and sand mining should be treated as water-quality issues, not only as morphological issues. Once the hydrological and hydraulic base state changes, the self-purification capacity and pollution distribution of the entire system change as well. Protecting surface water quality in the Mekong Delta

therefore requires coordinated action across internal pollution control, river-resource management, adaptive operation of hydraulic works, channel-morphology surveillance, and interregional information sharing with adjacent lower Dong Nai-Sai Gon and coastal receiving-water monitoring systems.

4. Conclusions

Surface water quality in the Vietnamese Mekong Delta shows strong spatial and temporal variability, with WQI ranging from 2.376 to 85.314 and a regional mean of about 47.486. This indicates that many monitored locations remain in the fair-to-poor range, especially in coastal zones, peri-urban reaches, aquaculture-intensive areas, and poorly flushed canal networks.

Pollution sources are the direct drivers of water-quality deterioration. Domestic and industrial wastewater dominate along urban axes and estuaries, whereas agriculture and aquaculture are more prominent in inner-delta and coastal settings; atmospheric deposition and the river-sea interface create distinctive pollution signatures in some estuarine areas.

Altered flow regimes are the main risk amplifier. Low flows, salinity intrusion, channel-bed lowering, and hydraulic-structure operation reduce dilution capacity, increase water residence time, and promote pollutant remobilization. Consequently, the same pollution load can produce stronger WQI decline when the assimilative capacity of the river-canal system has been reduced.

Management should therefore be prioritized by subregion and season: wastewater and industrial-effluent control for urban-estuarine corridors; aquaculture and agricultural runoff control for the Ca Mau Peninsula and inner-delta canals; WQI-based sluice operation for salinity-control systems; and channel-morphology monitoring for reaches affected by bed incision and sediment depletion.

The main limitation of this study is that it synthesizes monitoring, source-apportionment, and published/project evidence rather than recalculating a fully coupled hydrodynamic-

water-quality model from raw data. Differences in monitoring frequency, analytical protocols, and spatial density among datasets may affect comparability, and uncertainty bands for WQI or source-apportionment outputs were not newly estimated in this revision. In addition, the lower Dong Nai-Sai Gon area shown in Figure 1 is retained only as adjacent hydrological and source-receiving context; it is not treated as a Mekong Delta subregion in WQI aggregation or VMD-specific interpretation.

Future research should prioritize coupled hydrodynamic-salinity-water-quality forecasting, assimilation of high-frequency sensor and remote-sensing data, uncertainty quantification for WQI and source-apportionment outputs, and testing of WQI-based operation rules for salinity-control infrastructure. These developments should be connected to WebGIS and real-time decision-support systems so that water-quality warnings can be issued by subregion, season, and water-use objective.

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Declaration of originality: *The author declares that this manuscript is her own research work; that its content has not been published elsewhere in the same form; and that all data and cited materials are truthful, clearly sourced, and free from any related conflict of interest.*

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